

Homework 2

Due: Tuesday 4/8 at 11:59pm

Building on the given code, complete the problems below. (Remember to open the project by unzipping the file and clicking on `hw2.code-workspace`. Submit them as a .zip file of the directory and submitted through Google Classroom. Name your file based on your name or username so that it's clearly yours; my submission might be called `dbunde.zip`, `dbundeHW2.zip`, `david.zip`, `bunde.zip`, etc.

1. Consider a `Car` class with the following methods:

- A constructor that takes the initial value of the car (as in integer in US dollars). The newly-created car should initially be going at speed 0.
- A method `accelerate`, which takes an integer (assumed to be positive) and increases the current speed of the car by this many miles per hour. The method should return the car's current speed (after the accelerate operation).
- A method `crash`, which is called when the car gets into an accident. This is only valid if the car has a positive speed; if the car's speed is 0, the method should throw an `UnsupportedOperationException`. The method changes the car's speed to zero. The car's value also drops by 20 times the speed at which the car was driving. If this would make the car's value negative, it becomes 0 instead.
- A method `value` returns the current value of the car.

You will be writing test cases for this class. Note that you do NOT need to actually implement the class; you are only writing the test cases for it. (Though you're welcome to implement it if you'd like to do so since this will make it easier to check your test cases.)

- (a) (6 points) Write stub code for this class. (Methods with the right signatures that compile.) Again, you do NOT need to fully implement the methods.
 - (b) (18 points) Write JUnit test cases for this class.
2. (1 point) At the top of the file with your test cases, include a comment with a statement about your collaboration (if any). This should either be "I didn't work with anyone" or a description of who you talked with about the assignment and what you talked about. (For example: "David helped me create the JUnit file, X and I talked about testing accelerate.")